PROJECT PROGRESS REPORT

(Progress Report no. 2)

1. Student Information

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Enrollment Number	25IC03CA053
Program/Branch	Master Of Computer Applications
Semester	3
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2. Project Details

Type	Research Project
Department/Domain	MCA CA(DS/ML) / AI & ML
Mentor Name	Srikant Singh
Start Date	08-06-2026
End Date	03-10-2026

3. Work Progress Summary

a) Project Title:

- OpenCV - Based System for Crack Identification and Width Estimation

b) Requirement analysis:

The proposed system aims to automatically detect surface cracks and estimate their width using OpenCV image processing techniques. Traditional crack inspection is performed manually, which is time-consuming, subjective, and prone to human error.

The system should be capable of processing images captured from cameras or drones, identifying crack regions accurately, and estimating crack width in real time. It should provide reliable results with minimal human intervention and support different lighting conditions and crack patterns. The system should also generate visual outputs highlighting detected cracks along with their measured width for further analysis.

c) Feasibility study:

Technical Feasibility

- The project is technically feasible because it uses open-source technologies such as Python and OpenCV.
- Standard computers with moderate processing power are sufficient to execute image processing algorithms.

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Economic Feasibility

- The project has a low implementation cost since all required software tools are freely available.
- No expensive hardware or commercial software licenses are required.

Operational Feasibility

- The system is easy to operate and can assist engineers and inspectors in performing faster and more accurate crack analysis.
- It reduces manual effort and improves inspection efficiency.

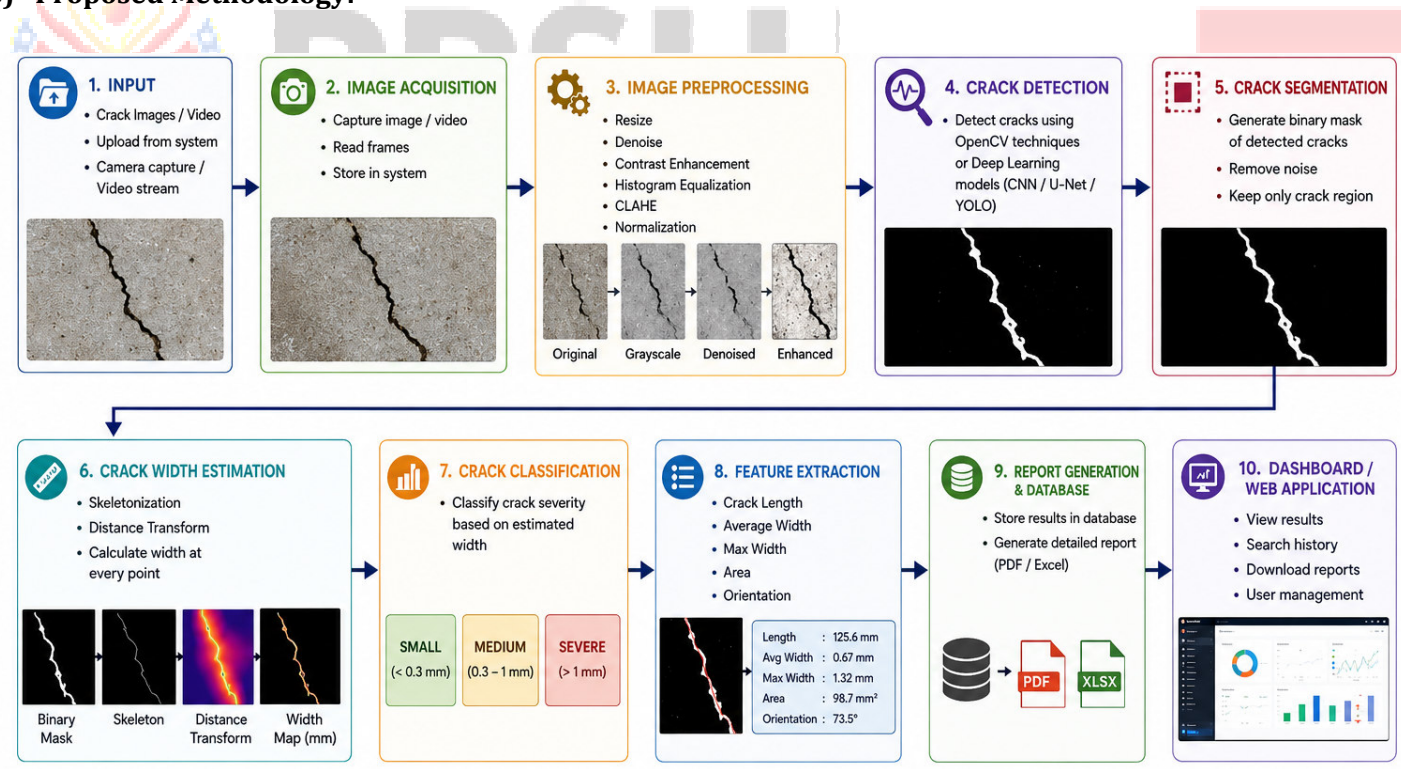
Schedule Feasibility

- The project can be completed within the allotted academic timeline by following phases such as literature review, system design, implementation, testing, evaluation, and documentation.

d) Technology Selection:

Component	Technology
Programming Language	Python
Image Processing	OpenCV
Numerical Computing	NumPy
Data Visualization	Matplotlib
Image Handling	Pillow
Development Environment	Visual Studio Code
Database Store Data	MongoDB
Operating System	Windows 10/11

e) Proposed Methodology:



4. Tasks Accomplished

- Conducted an extensive literature review on crack detection and width estimation techniques using computer vision.
- Analyzed project requirements and finalized the project objectives and scope.
- Selected the technology stack, including Python, OpenCV, NumPy, and Matplotlib, for system development.
- Designed the overall system architecture and processing workflow for automated crack identification.
- Collected and organized sample crack images for dataset preparation and testing.
- Implemented image preprocessing techniques such as grayscale conversion, image resizing, Gaussian filtering, and thresholding.
- Developed the crack detection module using OpenCV image processing algorithms, including edge detection and contour extraction.
- Designed the crack width estimation approach based on pixel measurements and image calibration.
- Implemented crack severity classification based on the estimated crack width.
- Performed initial testing of the detection pipeline and verified the accuracy of crack identification using sample images.
- Prepared the project workflow, methodology, and technical documentation for the progress report and presentation.

5. Tasks in Progress / Next Plan

- System analysis
- database design
- module planning
- development environment setup

6. Learnings / Skills Acquired

- Gained practical knowledge of Computer Vision and Digital Image Processing concepts.
- Learned to use the OpenCV library for image preprocessing, edge detection, contour detection, and crack segmentation.
- Improved programming skills in Python by implementing image processing algorithms.
- Acquired experience with NumPy and Matplotlib for image manipulation, numerical computation, and result visualization.
- Learned techniques for crack width estimation using pixel-based measurements and image calibration.
- Understood the design and implementation of a complete computer vision workflow, from image acquisition to crack severity classification.
- Developed problem-solving and debugging skills while testing and improving the crack detection pipeline.
- Enhanced knowledge of research methodology, technical documentation, and project presentation.
- Improved teamwork, time management, and project planning skills during the development process.
- Gained experience in evaluating system performance and interpreting experimental results for real-world structural inspection applications.

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References

- Bradski, G., & Kaehler, A. (2008). *Learning OpenCV: Computer vision with the OpenCV library*. " O'Reilly Media, Inc.".
- Pham, M. V., Ha, Y. S., & Kim, Y. T. (2023). Automatic detection and measurement of ground crack propagation using deep learning networks and an image processing technique. *Measurement*, *215*, 112832.
- Yang, Y. S., Wu, C. L., Hsu, T. T., Yang, H. C., Lu, H. J., & Chang, C. C. (2018). Image analysis method for crack distribution and width estimation for reinforced concrete structures. *Automation in Construction*, *91*, 120-132.
- Oh, J., Matsumoto, Y., & Arai, K. (2023). Development of Crack Detection and Crack Length Calculation Method using Image Processing. *International Journal of Advanced Computer Science & Applications*, *14*(12), 330.

Evaluation (To be filled by Mentor)

Criteria	Bloom's Level	Excellent (20)	Good (15)	Average (10)	Poor (5)	Marks Obtained (Out of 20)
Problem Definition	L4–Analyze	Clearly defined, justified real-world problem	Defined but lacks depth	Vague problem	No clear problem	
Project Title Relevance	L2–Understand	Highly precise and aligned	Mostly relevant	Partially relevant	Misleading	
Scope of Work	L5–Evaluate	Well-defined scope, deliverables, constraints	Defined with minor gaps	Broad/unclear	No scope	
Related Study	L4–Analyze	3–5 sources analyzed critically	Sources listed	Minimal references	No study	
Clarity & Presentation	L3–Apply	Structured, professional	Minor issues	Some confusion	Poor	
Total Marks						

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Evaluation

Student Signature	Mentor (Company) Signature

Mentor (Institute) Remarks

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